

Model Diagnostics and Visualization

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Table of contents

Introduction	1
Setup	2
Load Gu–Wahba example data	2
Fit the GAM	2
Model overview	3
Smooth estimates	4
Plotting smooth estimates with true functions	4
gratia’s built-in draw	6
Partial residuals	7
Derivatives	9
Plotting derivatives	9
Model diagnostics with appraise	11
Posterior simulation	12
Smooth samples (spaghetti plot)	12
Fitted samples	14
Concurvity	14
Basis dimension check	15
Summary	15

Introduction

This vignette demonstrates model diagnostics and visualization using **mgcv** and **gratia** in R. We fit the same Gu & Wahba (1991) example as the Julia vignette and compare diagnostic outputs.

Setup

```
library(mgcv)
```

Loading required package: nlme

This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.

```
library(gratia)
```

Load Gu–Wahba example data

```
dat <- read.csv("../data.csv")
n <- nrow(dat)
x0 <- dat$x0; x1 <- dat$x1; x2 <- dat$x2; x3 <- dat$x3
y <- dat$y

f0 <- function(x) 2 * sin(pi * x)
f1 <- function(x) exp(2 * x)
f2 <- function(x) 0.2 * x^11 * (10 * (1 - x))^6 + 10 * (10 * x)^3 * (1 - x)^10
f3 <- function(x) rep(0, length(x))

head(dat)
```

	y	x0	x1	x2	x3
1	2.992645	0.9148060	0.02270001	0.9090475	0.4018804
2	4.697172	0.9370754	0.51323953	0.8999248	0.4322142
3	13.935789	0.2861395	0.63072615	0.1923493	0.6636044
4	5.713290	0.8304476	0.41877162	0.5322903	0.1823693
5	7.634074	0.6417455	0.87926595	0.5221247	0.8383388
6	9.800107	0.5190959	0.10798707	0.1603357	0.9173730

Fit the GAM

```
m <- gam(y ~ s(x0, k = 10, bs = "cr") + s(x1, k = 10, bs = "cr") +
          s(x2, k = 10, bs = "cr") + s(x3, k = 10, bs = "cr"),
         data = dat, method = "REML")
summary(m)
```

Family: gaussian
Link function: identity

Formula:

```
y ~ s(x0, k = 10, bs = "cr") + s(x1, k = 10, bs = "cr") + s(x2,  
  k = 10, bs = "cr") + s(x3, k = 10, bs = "cr")
```

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.4951	0.1051	71.34	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(x0)	3.426	4.236	8.946	7.47e-07 ***
s(x1)	3.199	3.968	67.745	< 2e-16 ***
s(x2)	7.834	8.631	68.148	< 2e-16 ***
s(x3)	1.887	2.360	2.711	0.0578 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.685 Deviance explained = 69.8%
-REML = 881.96 Scale est. = 4.4158 n = 400

Model overview

gratia's overview provides a tidy summary of all smooth terms.

```
overview(m)
```

Generalized Additive Model with 4 terms

	term	type	k	edf	statistic	p.value
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<chr>
1	s(x0)	CRS	9	3.43	8.95	< 0.001
2	s(x1)	CRS	9	3.20	67.7	< 0.001
3	s(x2)	CRS	9	7.83	68.1	< 0.001
4	s(x3)	CRS	9	1.89	2.71	0.057796

Smooth estimates

`smooth_estimates` evaluates each smooth on a regular grid with pointwise standard errors.

```
se <- smooth_estimates(m, n = 200)
head(se)
```

```
# A tibble: 6 x 9
  .smooth .type .by .estimate .se x0 x1 x2 x3
  <chr>   <chr> <chr>   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 s(x0)   CRS <NA>   -1.32 0.392 0.000239 NA NA NA
2 s(x0)   CRS <NA>   -1.29 0.379 0.00525 NA NA NA
3 s(x0)   CRS <NA>   -1.25 0.366 0.0103 NA NA NA
4 s(x0)   CRS <NA>   -1.21 0.353 0.0153 NA NA NA
5 s(x0)   CRS <NA>   -1.18 0.341 0.0203 NA NA NA
6 s(x0)   CRS <NA>   -1.14 0.329 0.0253 NA NA NA
```

Plotting smooth estimates with true functions

```
true_fns <- list(f0, f1, f2, f3)
x_vars <- c("x0", "x1", "x2", "x3")
fn_labels <- c(
  expression(f[0](x[0]) == 2*sin(pi*x)),
  expression(f[1](x[1]) == exp(2*x)),
  expression(f[2](x[2]) ~ "Gu-Wahba"),
  expression(f[3](x[3]) == 0)
)

par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
for (i in 1:4) {
  se_i <- smooth_estimates(m, select = paste0("s(", x_vars[i], ")"), n = 200)
  x_grid <- se_i[[x_vars[i]]]
  est <- se_i[[".estimate"]]
  se_vals <- se_i[[".se"]]

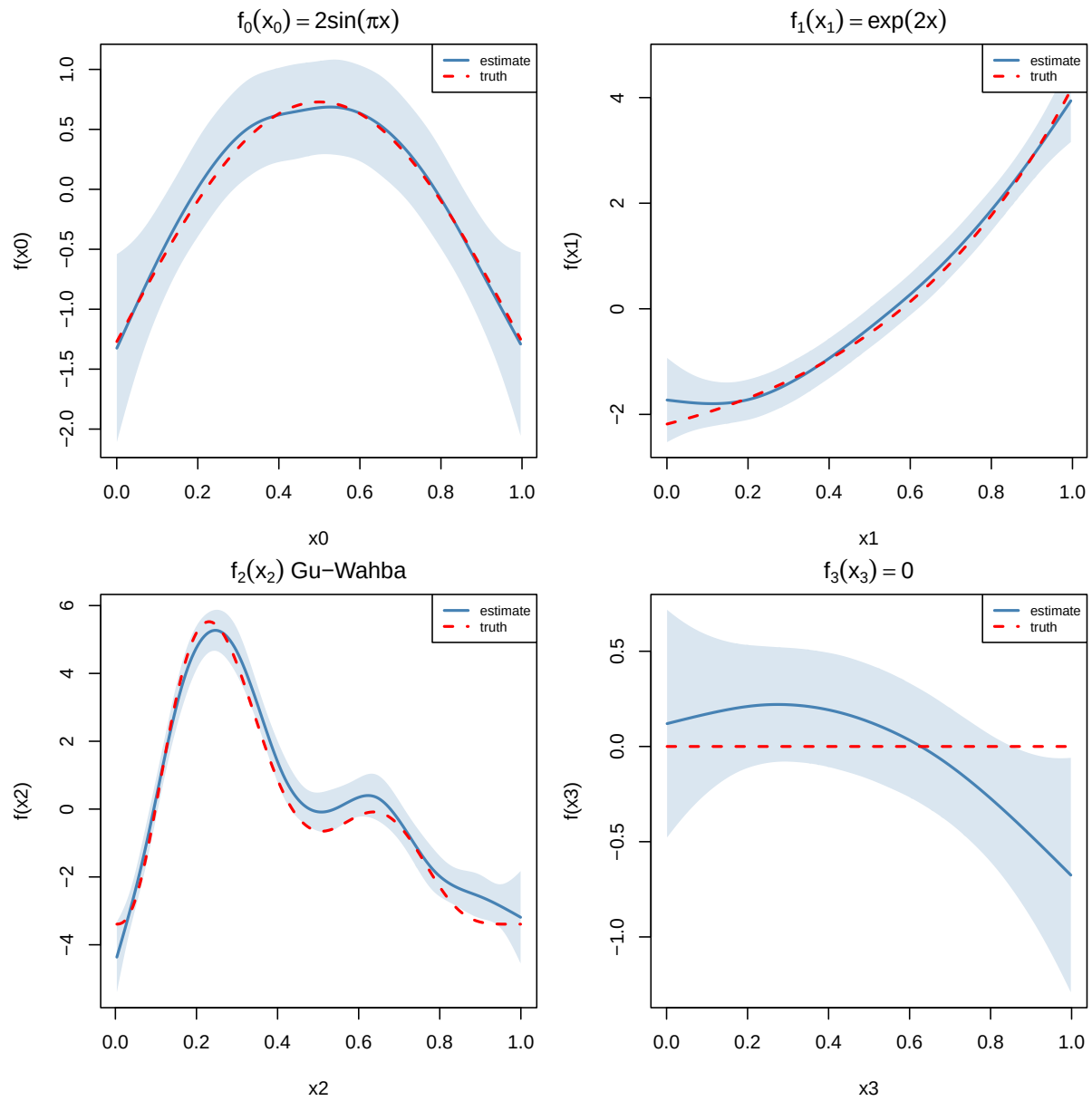
  f_grid <- true_fns[[i]](x_grid)
  f_grid <- f_grid - mean(f_grid)

  ylims <- range(c(est - 2 * se_vals, est + 2 * se_vals, f_grid))
  plot(x_grid, est, type = "l", lwd = 2, col = "steelblue",
```

```

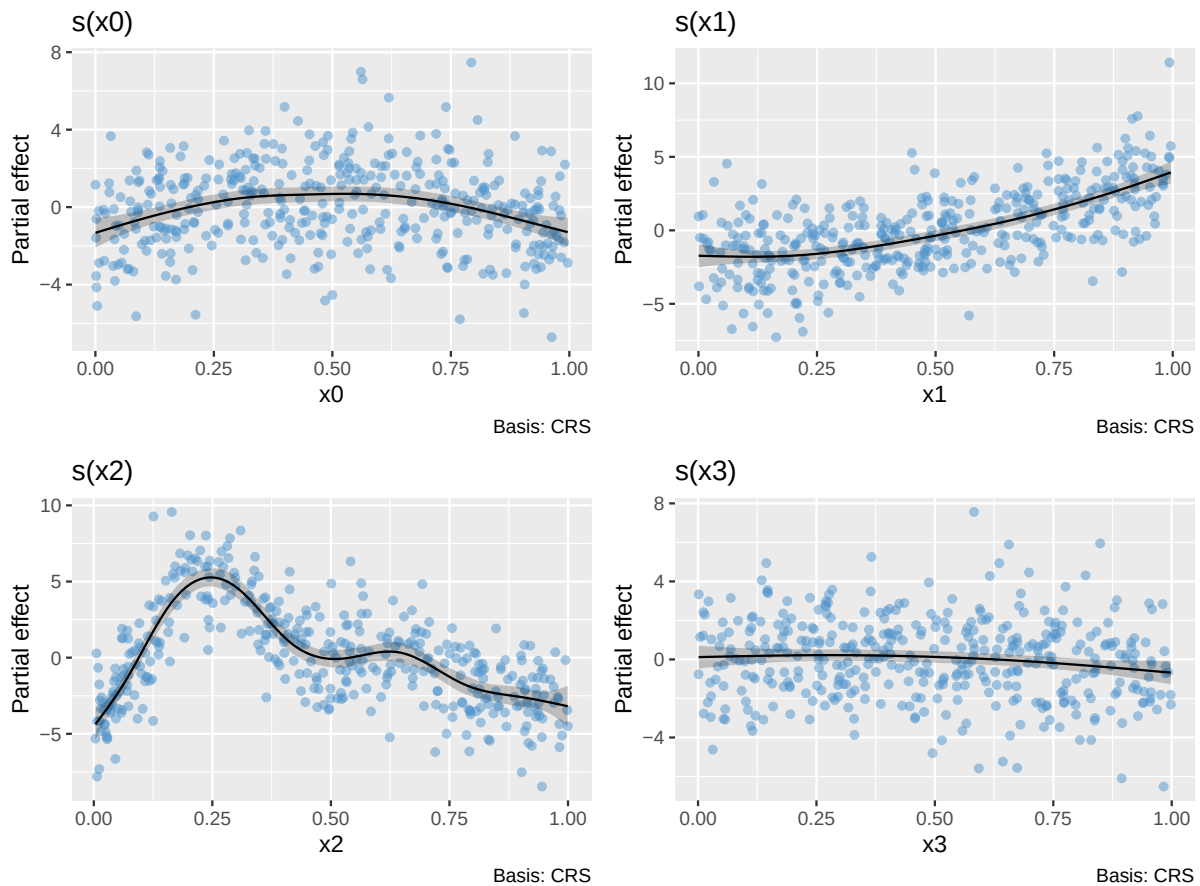
        ylim = ylims, xlab = x_vars[i], ylab = paste0("f(", x_vars[i], ")"),
        main = fn_labels[i])
    polygon(c(x_grid, rev(x_grid)),
            c(est - 2 * se_vals, rev(est + 2 * se_vals)),
            col = adjustcolor("steelblue", alpha.f = 0.2), border = NA)
    lines(x_grid, f_grid, lty = 2, lwd = 2, col = "red")
    legend("topright", c("estimate", "truth"), lty = c(1, 2),
           col = c("steelblue", "red"), lwd = 2, cex = 0.7, bg = "white")
}

```



gratia's built-in draw

```
draw(m, residuals = TRUE, rug = FALSE)
```



Partial residuals

Partial residuals overlaid on smooth estimates show how well the smooth captures the data pattern.

```
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
x_data <- list(x0, x1, x2, x3)
pr <- partial_residuais(m)

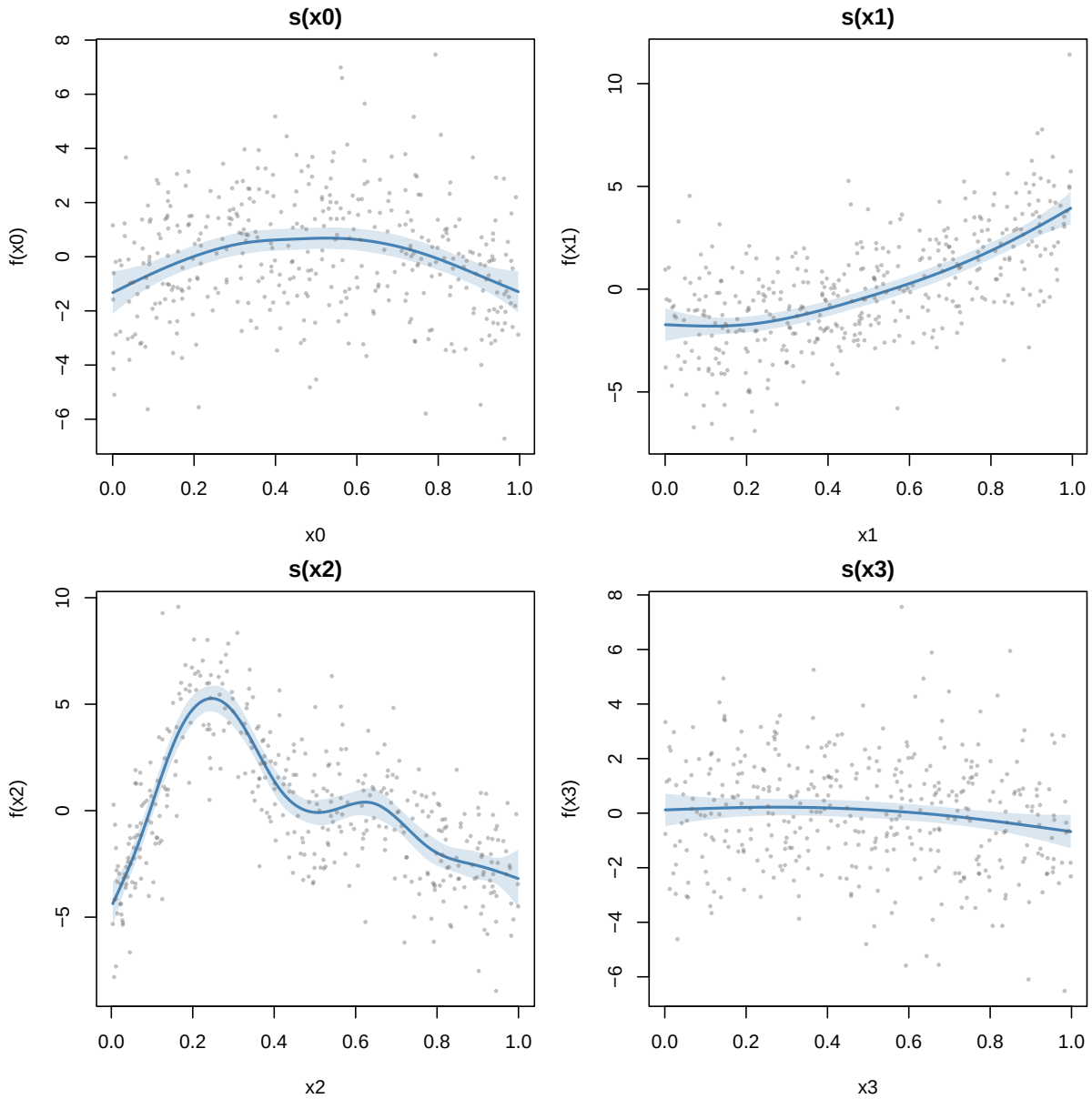
for (i in 1:4) {
  se_i <- smooth_estimates(m, select = paste0("s(", x_vars[i], ")"), n = 200)
  x_grid <- se_i[[x_vars[i]]]
  est <- se_i[[".estimate"]]
  se_vals <- se_i[[".se"]]

  plot(x_grid, est, type = "l", lwd = 2, col = "steelblue",
```

```

    xlab = x_vars[i], ylab = paste0("f(", x_vars[i], ")"),
    main = paste0("s(", x_vars[i], ")"),
    ylim = range(c(est - 2 * se_vals, est + 2 * se_vals, pr[[i]])))
  polygon(c(x_grid, rev(x_grid)),
    c(est - 2 * se_vals, rev(est + 2 * se_vals)),
    col = adjustcolor("steelblue", alpha.f = 0.2), border = NA)
  points(x_data[[i]], pr[[i]], pch = 16, cex = 0.5, col = adjustcolor("grey40", 0.4))
}

```



Derivatives

`derivatives` computes finite-difference derivatives with confidence intervals.

```
d <- derivatives(m, type = "central", n = 200)
head(d)
```

```
# A tibble: 6 x 12
  .smooth .by .fs .derivative .se .crit .lower_ci .upper_ci x0 x1
  <chr> <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 s(x0) <NA> <NA> 7.48 3.39 1.96 0.833 14.1 0.000239 NA
2 s(x0) <NA> <NA> 7.47 3.39 1.96 0.838 14.1 0.00525 NA
3 s(x0) <NA> <NA> 7.47 3.38 1.96 0.853 14.1 0.0103 NA
4 s(x0) <NA> <NA> 7.46 3.36 1.96 0.879 14.0 0.0153 NA
5 s(x0) <NA> <NA> 7.45 3.34 1.96 0.914 14.0 0.0203 NA
6 s(x0) <NA> <NA> 7.44 3.31 1.96 0.959 13.9 0.0253 NA
# i 2 more variables: x2 <dbl>, x3 <dbl>
```

Plotting derivatives

```
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
for (i in 1:4) {
  d_i <- derivatives(m, select = paste0("s(", x_vars[i], ")"),
                    type = "central", n = 200)
  x_grid <- d_i[[x_vars[i]]]
  deriv <- d_i[[".derivative"]]
  lower <- d_i[[".lower_ci"]]
  upper <- d_i[[".upper_ci"]]
  sig <- (lower > 0) | (upper < 0)

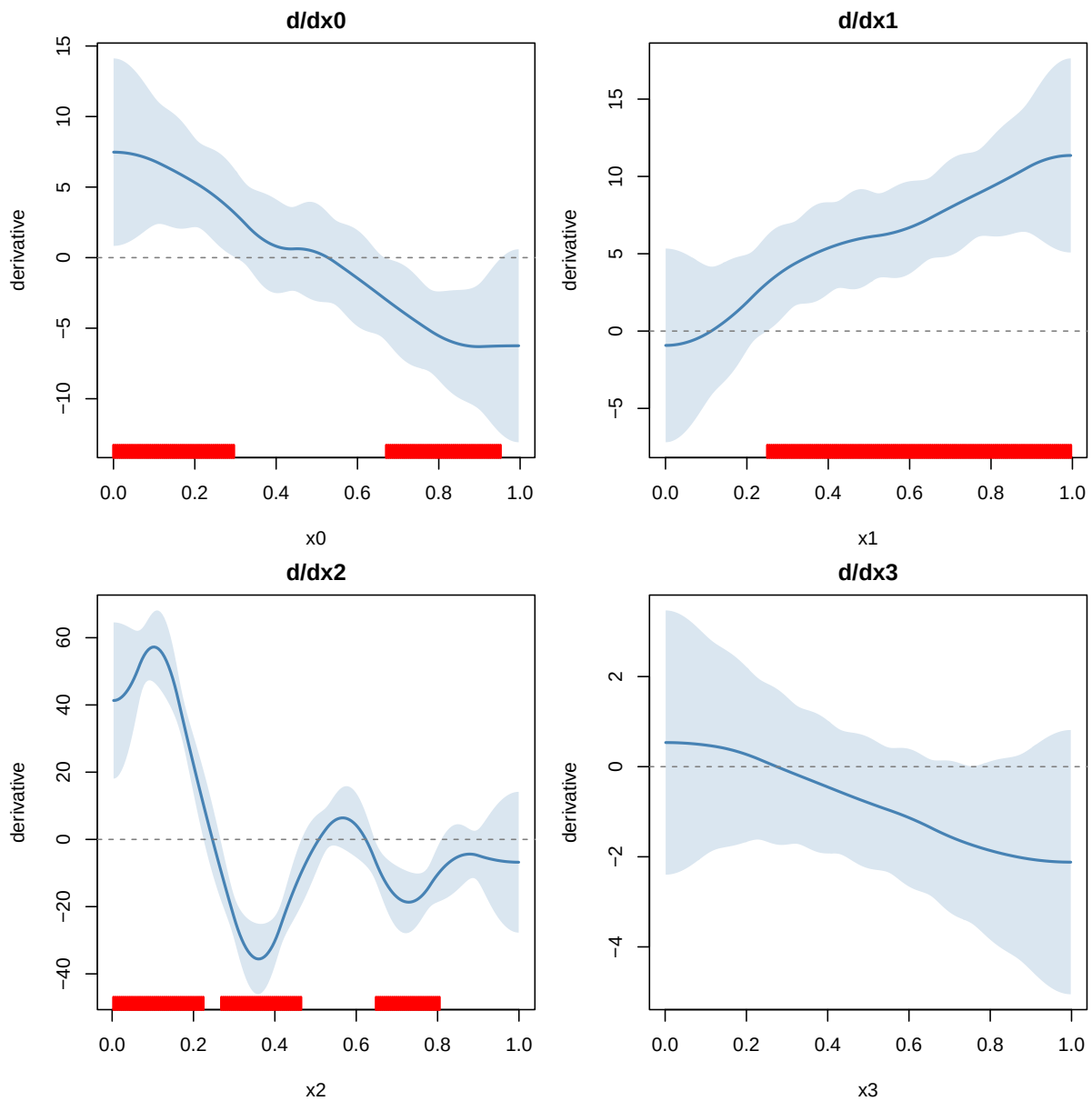
  ylims <- range(c(lower, upper))
  plot(x_grid, deriv, type = "l", lwd = 2, col = "steelblue",
       ylim = ylims, xlab = x_vars[i], ylab = "derivative",
       main = paste0("d/d", x_vars[i]))
  polygon(c(x_grid, rev(x_grid)),
         c(lower, rev(upper)),
         col = adjustcolor("steelblue", alpha.f = 0.2), border = NA)
  abline(h = 0, lty = 2, col = "grey50")

  if (any(sig)) {
```

```

    rug(x_grid[sig], col = "red", lwd = 2, side = 1)
  }
}

```



Second-order derivatives:

```

d2 <- derivatives(m, select = "s(x0)", order = 2, n = 200)
x_grid <- d2[["x0"]]

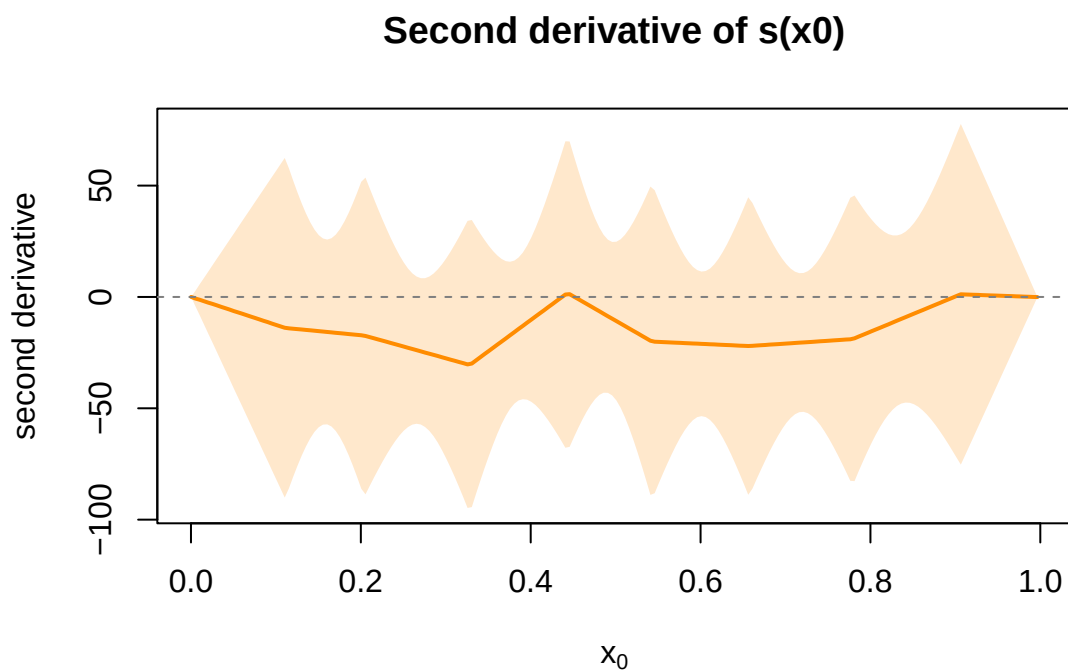
```

```

deriv <- d2[[".derivative"]]
lower <- d2[[".lower_ci"]]
upper <- d2[[".upper_ci"]]

plot(x_grid, deriv, type = "l", lwd = 2, col = "darkorange",
     ylim = range(c(lower, upper)),
     xlab = expression(x[0]), ylab = "second derivative",
     main = "Second derivative of s(x0)")
polygon(c(x_grid, rev(x_grid)),
       c(lower, rev(upper)),
       col = adjustcolor("darkorange", alpha.f = 0.2), border = NA)
abline(h = 0, lty = 2, col = "grey50")

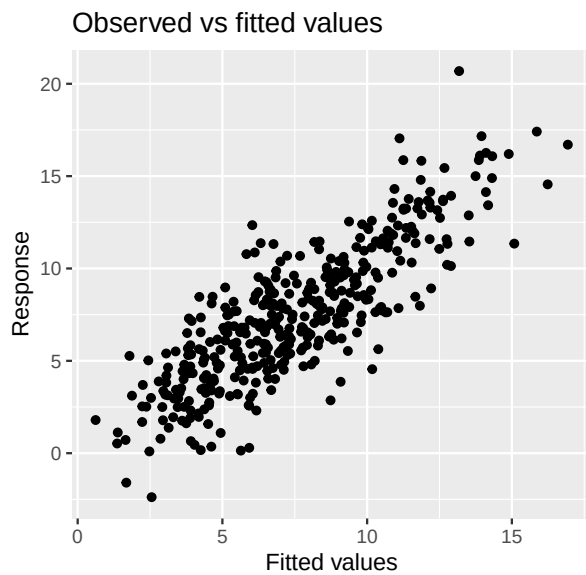
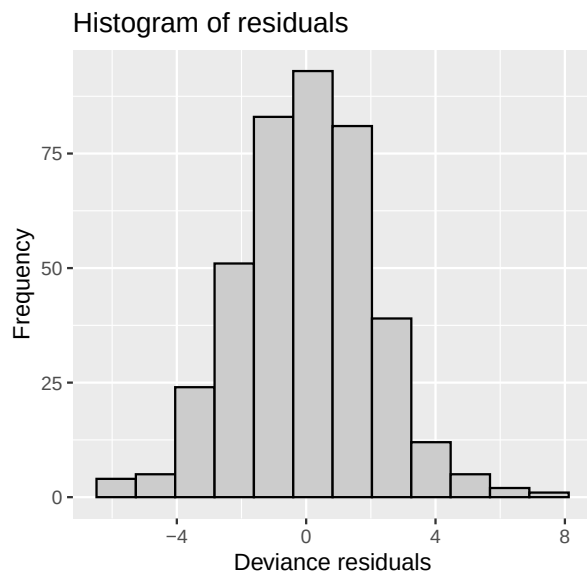
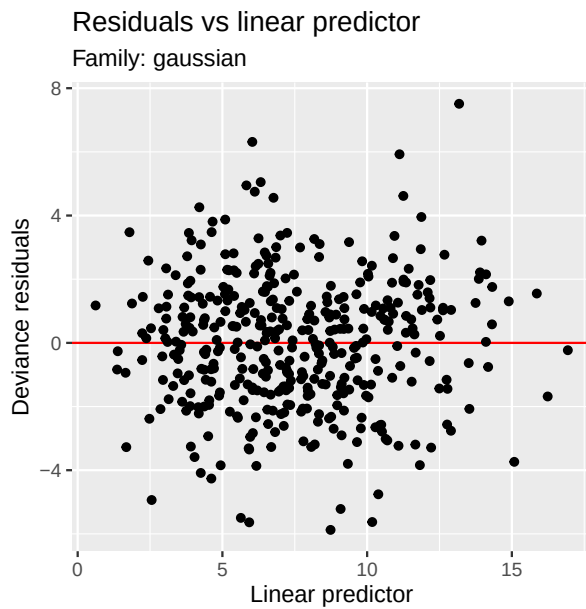
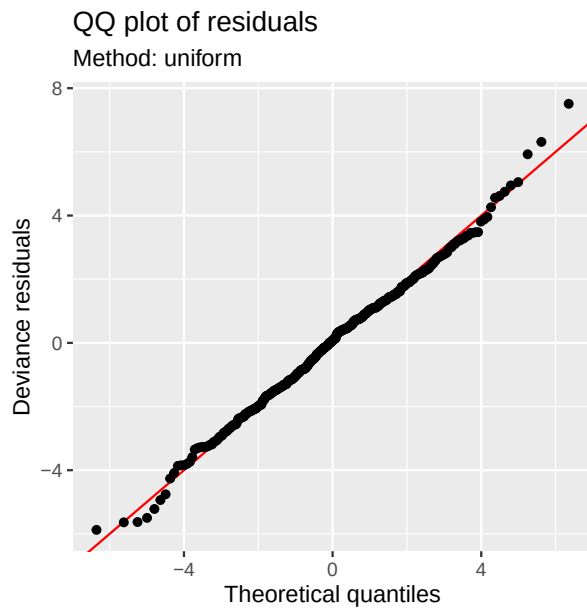
```



Model diagnostics with `appraise`

`appraise` produces the four standard diagnostic plots: QQ, residuals vs linear predictor, histogram, and observed vs fitted.

```
appraise(m)
```



Posterior simulation

Smooth samples (spaghetti plot)

`smooth_samples` draws posterior realizations of individual smooth functions.

```
ss <- smooth_samples(m, select = "s(x0)", n = 50, seed = 42)
```

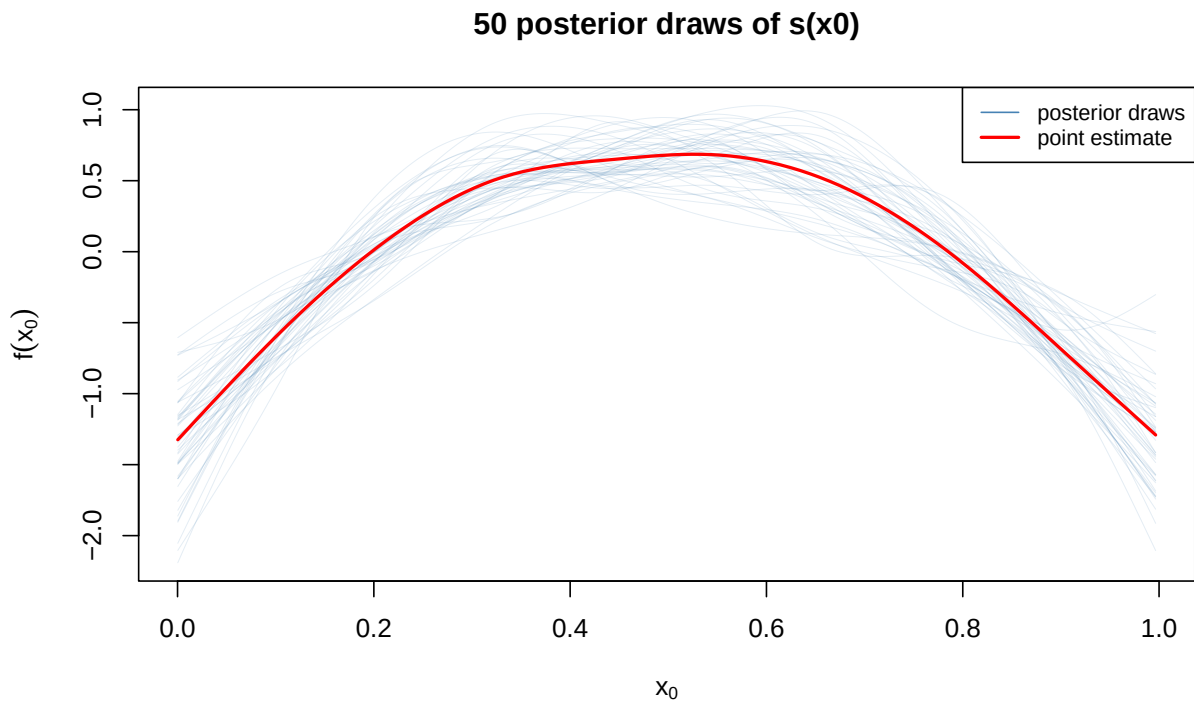
```

se_x0 <- smooth_estimates(m, select = "s(x0)", n = 100)
x_grid <- se_x0[["x0"]]
est <- se_x0[[".estimate"]]

draws <- unique(ss[[".draw"]])
ylims <- range(ss[[".value"]])

plot(NULL, xlim = range(x_grid), ylim = ylims,
      xlab = expression(x[0]), ylab = expression(f(x[0])),
      main = "50 posterior draws of s(x0)")
for (dr in draws) {
  idx <- ss[[".draw"]] == dr
  lines(ss[["x0"]][idx], ss[[".value"]][idx],
        col = adjustcolor("steelblue", 0.15), lwd = 0.5)
}
lines(x_grid, est, col = "red", lwd = 2)
legend("topright", c("posterior draws", "point estimate"),
      col = c("steelblue", "red"), lwd = c(1, 2), cex = 0.8)

```



Fitted samples

`fitted_samples` draws posterior samples of fitted values.

```
fs <- fitted_samples(m, n = 200, seed = 42)
cat("Fitted samples: ", nrow(fs), "rows x", length(unique(fs[[".draw"]])), "draws\n")
```

Fitted samples: 80000 rows x 200 draws

Concurvity

```
model_concurvity(m)
```

```
# A tibble: 15 x 3
  .type      .term .concurvity
  <chr>      <chr>    <dbl>
1 worst     para      2.37e-30
2 worst     s(x0)      1.28e- 1
3 worst     s(x1)      1.39e- 1
4 worst     s(x2)      1.32e- 1
5 worst     s(x3)      1.68e- 1
6 observed  para      2.37e-30
7 observed  s(x0)      4.14e- 2
8 observed  s(x1)      5.32e- 2
9 observed  s(x2)      5.35e- 2
10 observed s(x3)      5.79e- 2
11 estimate para      2.37e-30
12 estimate s(x0)      6.57e- 2
13 estimate s(x1)      7.44e- 2
14 estimate s(x2)      6.23e- 2
15 estimate s(x3)      7.25e- 2
```

Pairwise concurvity:

```
model_concurvity(m, pairwise = TRUE)
```

```
# A tibble: 75 x 4
  .type .term .with .concurvity
  <chr> <chr> <chr>    <dbl>
```

```

1 worst para para      1    e+ 0
2 worst para s(x0)    4.28e-31
3 worst para s(x1)    3.05e-31
4 worst para s(x2)    2.95e-31
5 worst para s(x3)    4.67e-31
6 worst s(x0) para    3.45e-31
7 worst s(x0) s(x0)    1    e+ 0
8 worst s(x0) s(x1)    6.60e- 2
9 worst s(x0) s(x2)    8.55e- 2
10 worst s(x0) s(x3)    5.08e- 2
# i 65 more rows

```

Basis dimension check

```
k.check(m)
```

```

      k'      edf  k-index p-value
s(x0)  9 3.426140 1.0475474 0.8275
s(x1)  9 3.199210 1.0197537 0.6175
s(x2)  9 7.833919 1.0422706 0.7975
s(x3)  9 1.886556 0.9756429 0.2825

```

Summary

Both GAM.jl and gratia provide equivalent diagnostic interfaces:

gratia (R)	GAM.jl (Julia)
overview()	overview()
smooth_estimates()	smooth_estimates()
derivatives()	derivatives()
appraise()	appraise()
posterior_samples()	posterior_samples()
fitted_samples()	fitted_samples()
smooth_samples()	smooth_samples()
partial_residuals()	partial_residuals()
model_concurvity()	concurvity()
k.check()	k_check()